HAT-P-12b

R-1413

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-12_210510` · created 2026-08-01T05:26:01+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459344.8123 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1611 +/- 0.0098
Transit depth (Rp/Rs)^2	2.6 +/- 0.32 [%]
Orbital Inclination (inc)	88.84 +/- 0.76
Airmass coefficient 1 (a1)	0.976 +/- 0.015
Airmass coefficient 2 (a2)	0.023 +/- 0.018
Scatter in the residuals of the lightcurve fit is	1.49 %
Best Comparison Star	#3 - [228, 361]
Optimal Aperture	3.24
Optimal Annulus	4.48
Transit Duration (day)	0.098 +/- 0.0033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1611 ± 0.0098 vs 0.1406 (deviation 2.09σ)
Duration fit vs published	0.098 d vs 0.0975 d (deviation 0.15σ)
Mid-time O–C	-2.3 min
Depth SNR	16.4
Residual scatter	1.49 %

Light curve

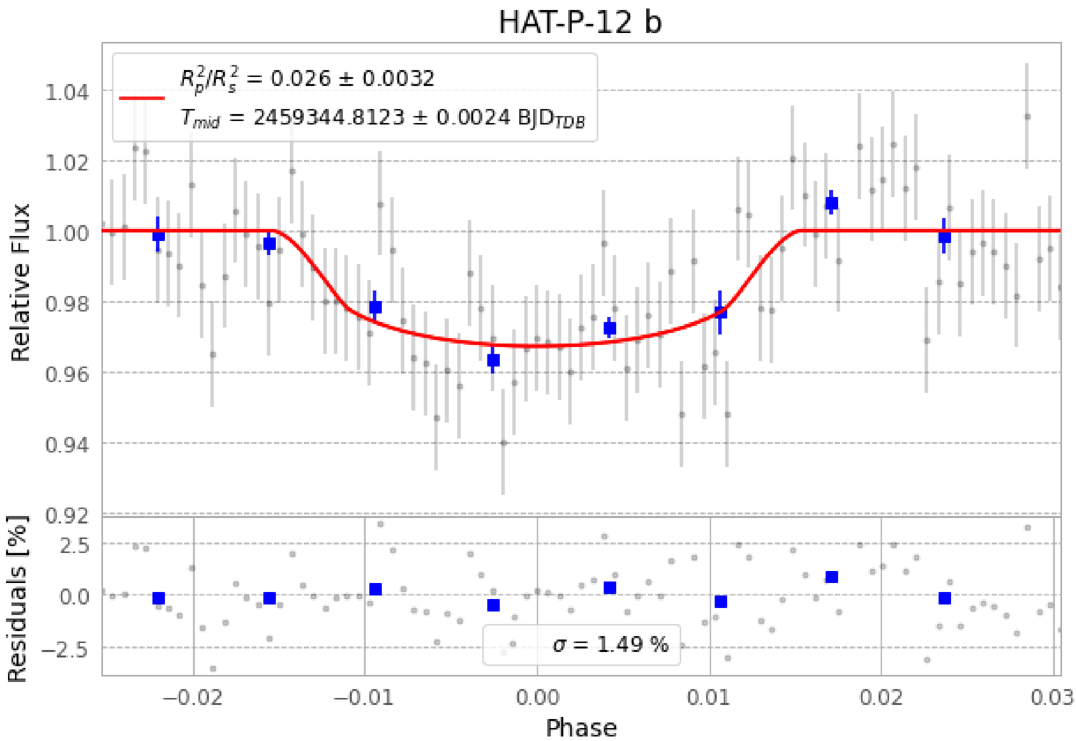

Figure 1 — FinalLightCurve_HAT-P-12 b_2021-05-09.png (SHA-256 f0797c59333b5cd9…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [391, 221], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 80ed07af297a6a63…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

87 FITS frames (57 MB), `HATP-12210510052712.FITS` → `HATP-12210510094512.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-210510.log (6ed91a3d9a95…), FinalLightCurve_HAT-P-12 b_2021-05-09.png (f0797c59333b…), FinalLightCurve_HAT-P-12 b_2021-05-09.csv (9e44b1060f93…)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)